

Qn 1	Answer	Marks
(a)(i)	$h = u_y t - \frac{1}{2} g t^2 = (12.4 \sin 36^\circ)(0.17) - \frac{1}{2}(9.81)(0.17)^2$ $= 1.10 \text{ m}$	[1]
(a)(ii)	The speed of the ball is given by $v = \sqrt{v_x^2 + v_y^2}$ $v_x = u_x = u \cos \theta = 12.4 \cos 36^\circ$ and $v_y = u_y + a_y t = (12.4 \sin 36^\circ) - (9.81)(0.17)$ $v = \sqrt{v_x^2 + v_y^2} = \sqrt{(10.03^2) + (5.62^2)} = 11.5 \text{ m s}^{-1}$ $\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{5.62}{10.03}\right) = 29.3^\circ$ Direction of the velocity = <u>29.3° above the horizontal</u> allow e.c.f. for \ deduct last A1 mark if student just determines angle without stating with respect to horizontal or vertical.	[1] [1] [1]
(b)	smooth curve with ball hitting wall below original, initial starting point must be tangent to the original path. Smooth asymmetrical curve showing rebound to ground with correct reflection at wall	[1] [1]
	Max. Marks	6

2020

HWA CHONG INSTITUTION (COLLEGE SECTION)

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